

Advanced Ultrasound Signal Processing on GPUs

Intro & CUDA Update

Marcin Lewandowski, Piotr Jarosik, Piotr Karwat

SUPPORT: Damian Cacko, Mateusz Walczak

**LAB
4US**

DISCLAIMER

- Content and works included in this educational materials/presentations are subject to the rules resulting from the Polish regulations on copyright and related rights. Downloading to a computer and copying of all or part of educational materials/presentations is permitted only for non-commercial use under the terms of the CC BY-NC-ND 4.0 licence.
- Reproduction, processing in whole or in part and any type of use going beyond that referred to in the preceding sentence requires the written permission of the creator/entity holding the copyright to the educational material/presentation.
- In the educational material/presentation, content and works copyrighted by others have been used within the scope of permitted use of protected works — such content and works have been marked as such. If you nevertheless notice a copyright infringement, please inform us accordingly. In the event of an infringement, we will remove the relevant content immediately.
- Every effort has been made to ensure that all borrowed material has its source and/or author correctly stated.
- The information contained in the material has been prepared to the best knowledge of the authors and is taken from sources believed to be reliable, although the authors do not guarantee its accuracy or completeness.



TEAM / SPEAKERS

**Dr Marcin Lewandowski <marcin@us4us.eu> / us4us Ltd., Poland.**

Since receiving a Masters degree in Physics followed by a PhD in Electronic Engineering, Marcin Lewandowski has headed many projects in R&D, commercial product design and medical devices development and certification. He has also authored numerous publications in scientific journals on the medical and industrial applications of ultrasound. Over the course of 30+ years working in ultrasound, electronics and software development, Marcin has strived to apply his research expertise in projects with a strong potential for innovation and commercialization. Today, he balances his continued work in science with his role as CEO at us4us Ltd., who produce original ultrasound platforms for research, biomedical and industrial applications.

Novel ultrasound imaging solutions and GPU implementation have formed a key part of Marcin's work for several years. us4us products all employ software processing on GPUs for real-time access to raw ultrasound data, and garner interest from universities, research institutes and commercial clients looking for powerful, versatile and fully programmable imaging solutions.

**Piotr Jarosik <piotr.jarosik@us4us.eu> / us4us Ltd., Poland.**

Piotr Jarosik received his Bachelor's and Masters Degree in Computer Science at Faculty of Electronics and Information Technology, Warsaw University of Technology. Currently, he is a software architect at us4us and a PhD student at IPPT PAN. His research interests include machine learning and ultrasound data processing.

**Dr Piotr Karwat <piotr.karwat@us4us.eu> / us4us Ltd., Poland.**

Piotr Karwat studied at the Warsaw University of Technology, Faculty of Mechatronics, specializing in Biomedical Engineering, and earned his Master's degree in 2008. Since 2009, he has been working at the Institute of Fundamental Technological Research of the Polish Academy of Sciences (IPPT PAN), where in 2017 he defended his Ph.D. in Electronics with a dissertation entitled "Numerical reconstruction of the acoustic wave velocity field based on ultrasound signal analysis."

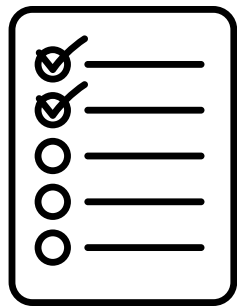
In 2019, he joined us4us ltd., where he continues his research and professional work. His expertise lies in the analysis and processing of ultrasound signals, particularly in the context of B-mode imaging, Doppler techniques, and quantitative ultrasound.

SUPPORT TEAM: Mateusz Walczak, Damian Cacko

THANK YOU!

PLAN

*... Simple, easy to remember.
- Jack Sparrow*



- Welcome & Intro
- “The hard core/code”
- Q&A / Discussion

Presentation	Instructor
0. Welcome & Introduction	Marcin Lewandowski
1. GPU Programming Refresher	Marcin Lewandowski, Piotr Jarosik
2. GPU Performance Guidelines and Patterns	Piotr Jarosik
3. CUDA in practice: ray-tracing	Piotr Karwat
4. Software Defined Ultrasound Architecture with GPU Processing	Piotr Jarosik

WELCOME!

>> <https://github.com/Lab4US/advanced-us-gpu-short-course> <<

- 1st edition of our upgraded/advanced GPU short-course on the IUS!
 - Previous basic GPU short-course still available:
 - >> <https://github.com/Lab4US/gpu-short-course>

NEW

- NVIDIA CUDA and AMD HIP
- Advanced GPU optimization
- Ultrasound system with GPU – system level optimization



**LAB
4US**

us4us[®]

INTRO & PRE-REQUISITE



IEEE IUS 2021

us4us[®]

IPPT PAN

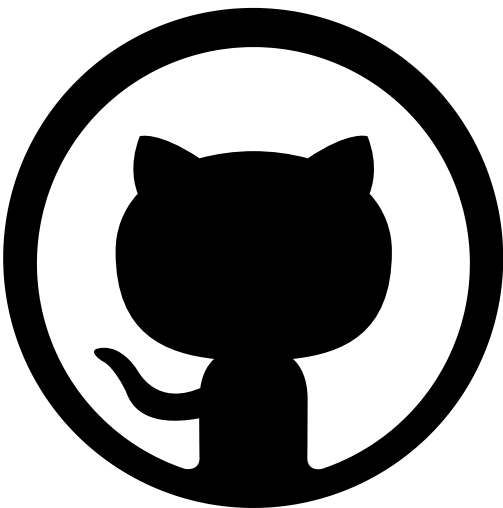
UNIVERSITY OF WATERLOO

2022 US-GPU short-course

A short-course: "Ultrasound Signal Processing with GPUs — Introduction to Parallel Programming" presented on the IEEE International Ultrasonics Symposium (IUS) in 2021/2022. [Authors: Marcin Lewandowski (us4us), Piotr Jarosik (us4us), Billy Yiu (LITMUS, UWaterloo)]

▶ Start watching

Ultrasound Signal Processing with GPUs – Introduction to Parallel Programming



<https://github.com/Lab4US/gpu-short-course>

IEEE IUS 2021

us4us[®]

IPPT PAN

UNIVERSITY OF WATERLOO

2022 US-GPU short-course

A short-course: "Ultrasound Signal Processing with GPUs — Introduction to Parallel Programming" presented on the IEEE International Ultrasonics Symposium (IUS) in 2021/2022. [Authors: Marcin Lewandowski (us4us), Piotr Jarosik (us4us), Billy Yiu (LITMUS, UWaterloo)]

▶ Start watching

Ultrasound Signal Processing with GPUs – Introduction to Parallel Programming

1-lecture-GPU-0-INTRO

21:07

2-lecture-GPU-1-CUDA

28:05

3-exercise-CUDA Progra...

54:03

4-lecture-GPU-2-CUDA-M...

25:45

5-exercise-CUDA Memor...

66:57

6-exercise-Performance ...

29:35

7-lecture-US Hardware an...

40:39

8-exercise-CUDA stream...

38:27

9-case-study-Color Doppl...

31:36

10-case-study-CESI platfo...

29:58

11-case-study-Shear Wav...

20:59

12-IUS-2022-SWE-DEMO

16:05

<https://vimeo.com/showcase/2022-us-gpu-short-course>

GPU @ us4us

2 DEMOS = 2X NVIDIA IGX + NVIDIA HOLOSCAN

NVIDIA Holoscan

is a multimodal computing platform that delivers the accelerated, full-stack infrastructure you need for scalable, software-defined, and real-time processing of streaming data running at the edge.

NVIDIA IGX



us4R-lite™

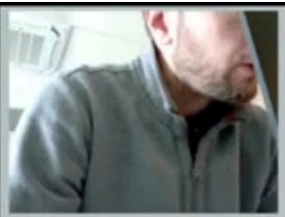
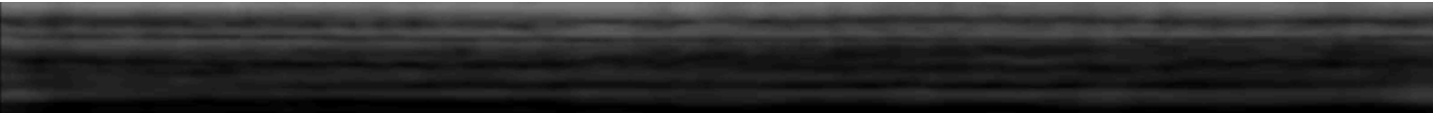


40Gb

Thanks NVIDIA for providing us
with the IGXs

- Our demos runs on NVIDIA IGX — a family is an enterprise-grade edge AI platform designed for safety-critical, real-time applications in industrial, medical, and robotics environments.
- Our open-source ARRUS SDK fully supports this platform (SoC ARM64).
- The us4R-lite ultrasound platform is connected using PCIe or 40Gb Ethernet (experimental).
- **us4R-lite integration with the NVIDIA HOLOSCAN opens new ways for a high performance edge computing.**

MR. MRAU – MULTIMODAL REASONING AGENT FOR ULTRASOUND



MRAU:

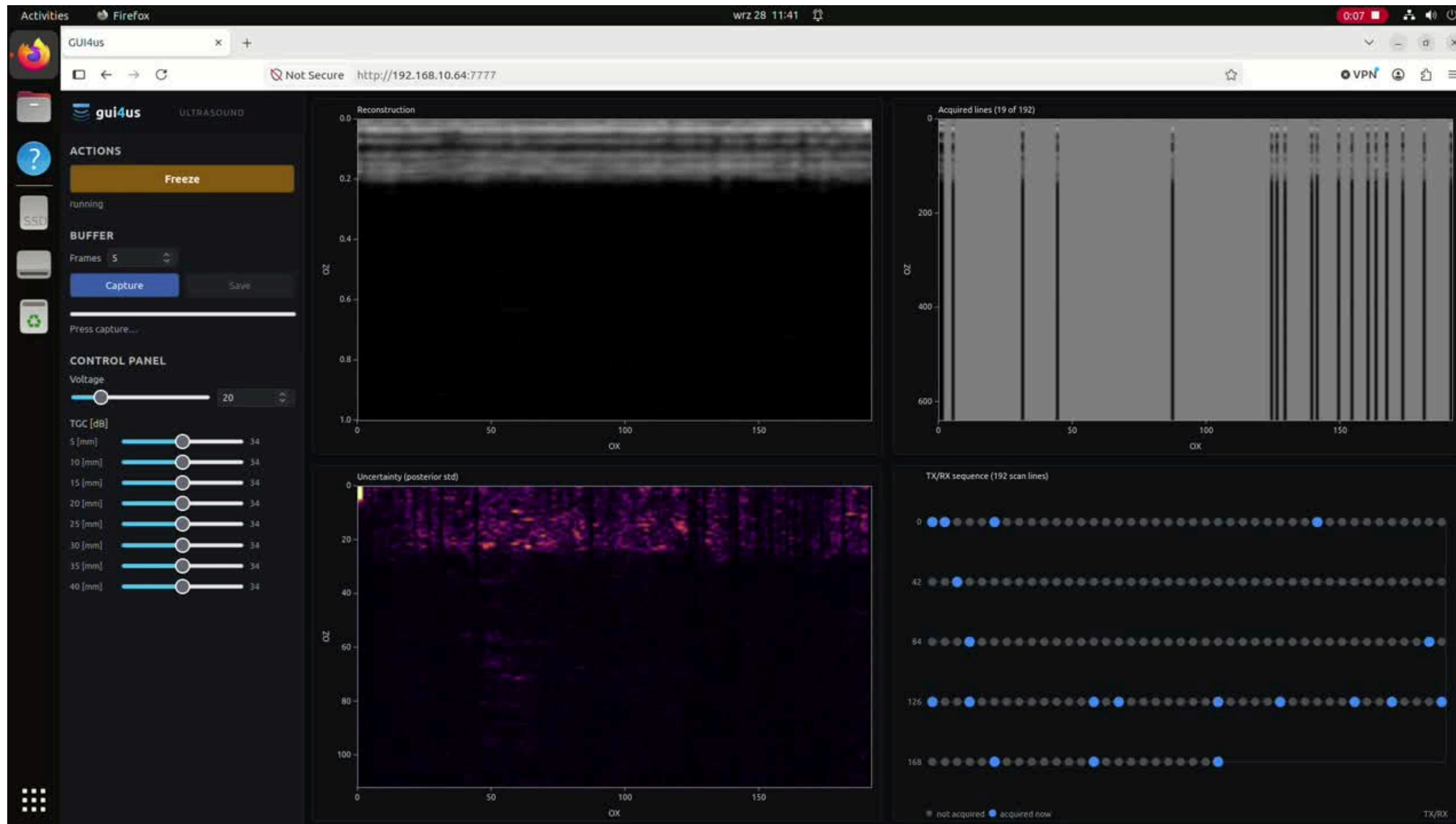
What are we looking for? carotid artery|
Your opinion:



Carotid artery is visible on the left side of the body.

Looking for: carotid artery · mic -60 dBFS · listening · REC 0:00 · F6 stop · F2 good · F4 bad · Collecting temporal context for the next?

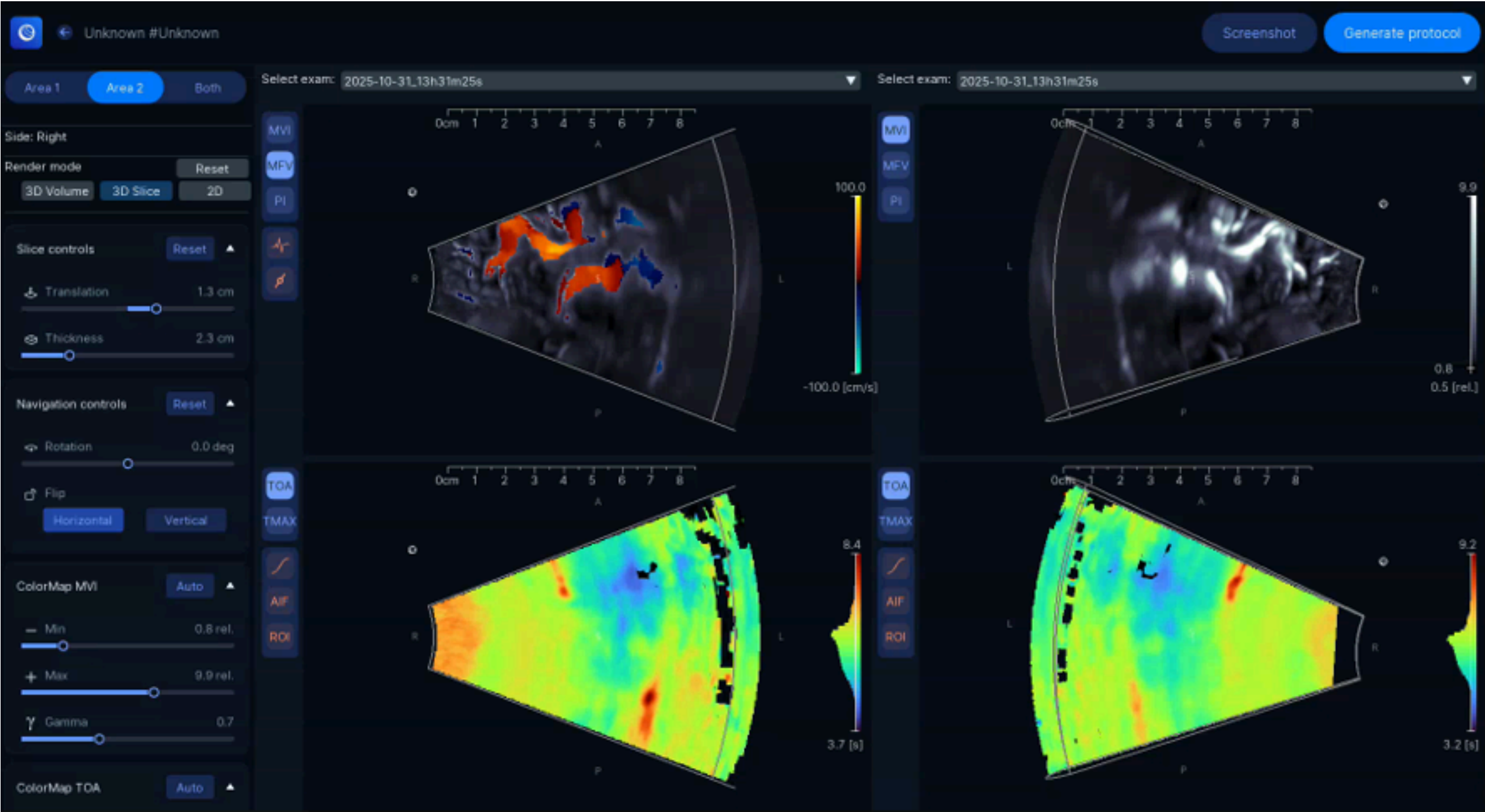
COGNITIVE ULTRASOUND + HARDWARE-IN-THE-LOOP



The grey area is not acquired at all!

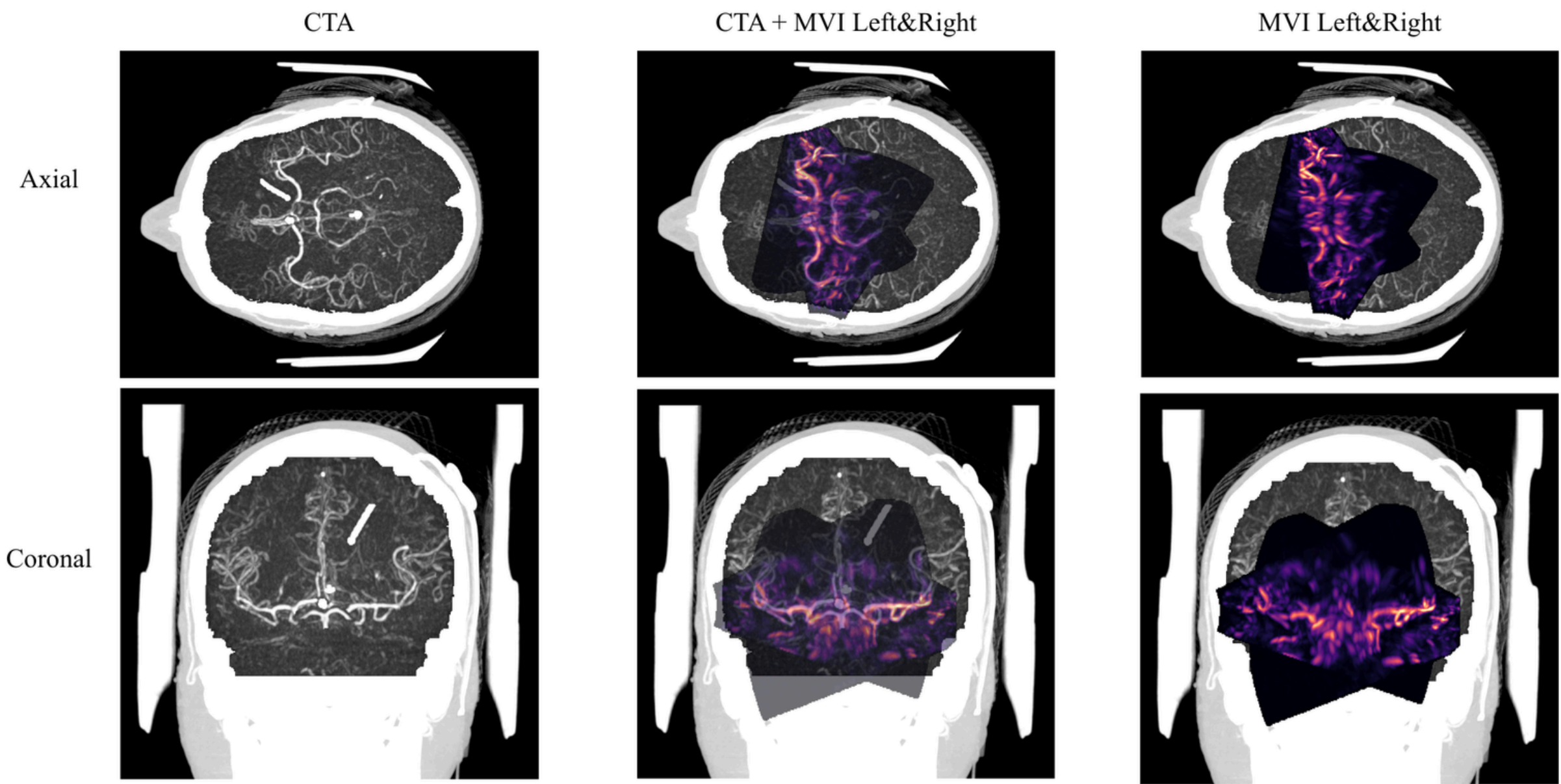
us4OEM™ in Medical Devices

FIRST CLINICAL SYSTEM FOR 3D BRAIN IMAGING



Source: materials provided by Resolve-Stroke, 2026

FLOW VALIDATED AGAINST THE CT



Dr Gakuba, Caen



Dr Szabo, Montpellier



Dr Kassem/Gory,
Nancy

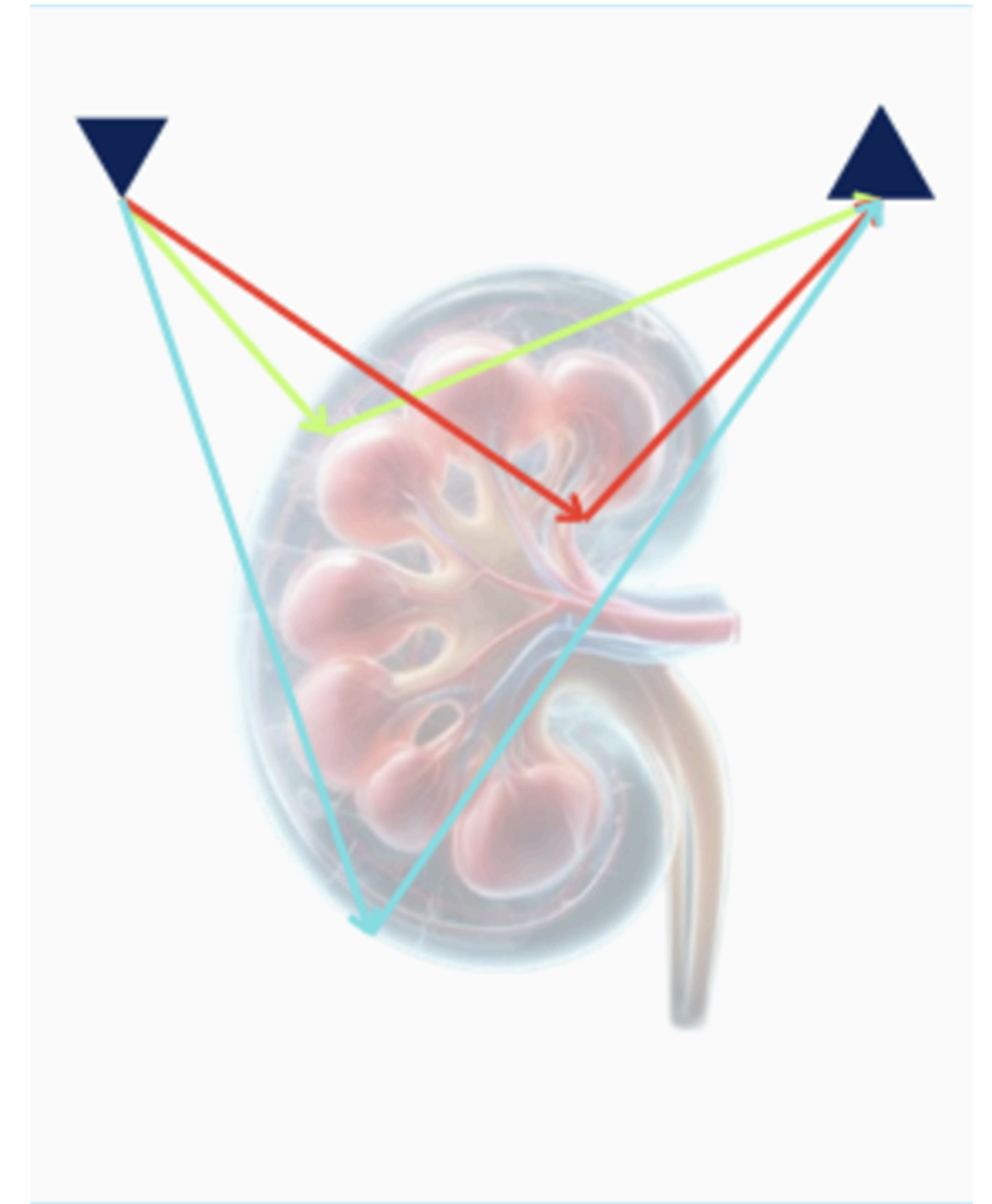


Dr Knieling, Erlangen

VORTEX-IMAGING: VOLUMETRIC ULTRASOUND, BUILT ON US4OEM



- Full-waveform inversion imaging
 - The whole volume acquired and solved at once.
- BENEFITS:
 - ~3 sec acquisition
 - ~5 min to DICOM
 - 0 radiation
 - Volumetric, quantitative, operator independent



Every echo kept. **Physics and GPU** compute do the rest.

PROBE

Matrix array Transducer Probe

Wide aperture, $\sim 12 \times 12$ cm · handheld with screen

ACQUIRE

us4OEM · us4us

Transmit pulsers and receive digitizers · full-aperture TX/RX · raw RF data

COMPUTE

Nvidia Jetson

Live positioning, coupling and stability feedback · encrypted upload

SOLVE

Cloud GPUs · FWI

Compute on cloud, on high-end Nvidia GPUs · 3D wave-equation reconstruction · volume returned as DICOM



Acquisition solved. Our effort goes into the physics.

us4OEM is the transmit and receive front end of every Vortex device. Everything above it is Vortex software.

0 1

Acquisition

Pulsers, receivers and digitizers in one module. Raw RF from every channel, exactly what FWI needs.

0 2

Software defined

Transmit schemes, sampling and the data path are written in code through the ARRUS API.

0 3

Scalable channels

OEM cards cascade. Channel count grows with the probe, from today's aperture to the next one.

0 4

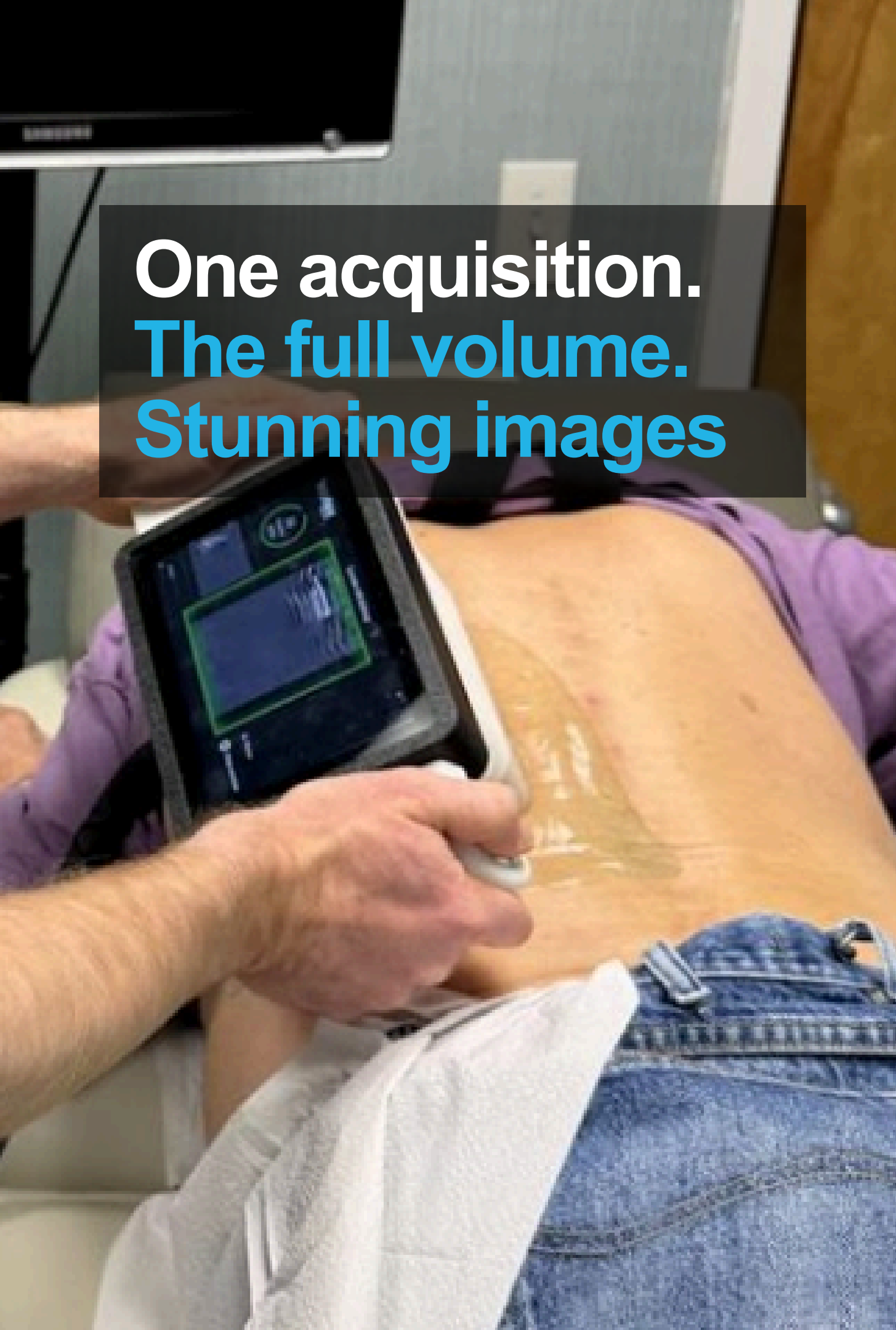
Reliable

ISO 13485 manufacturing, per-module test protocols and risk files that went straight into our 510(k).

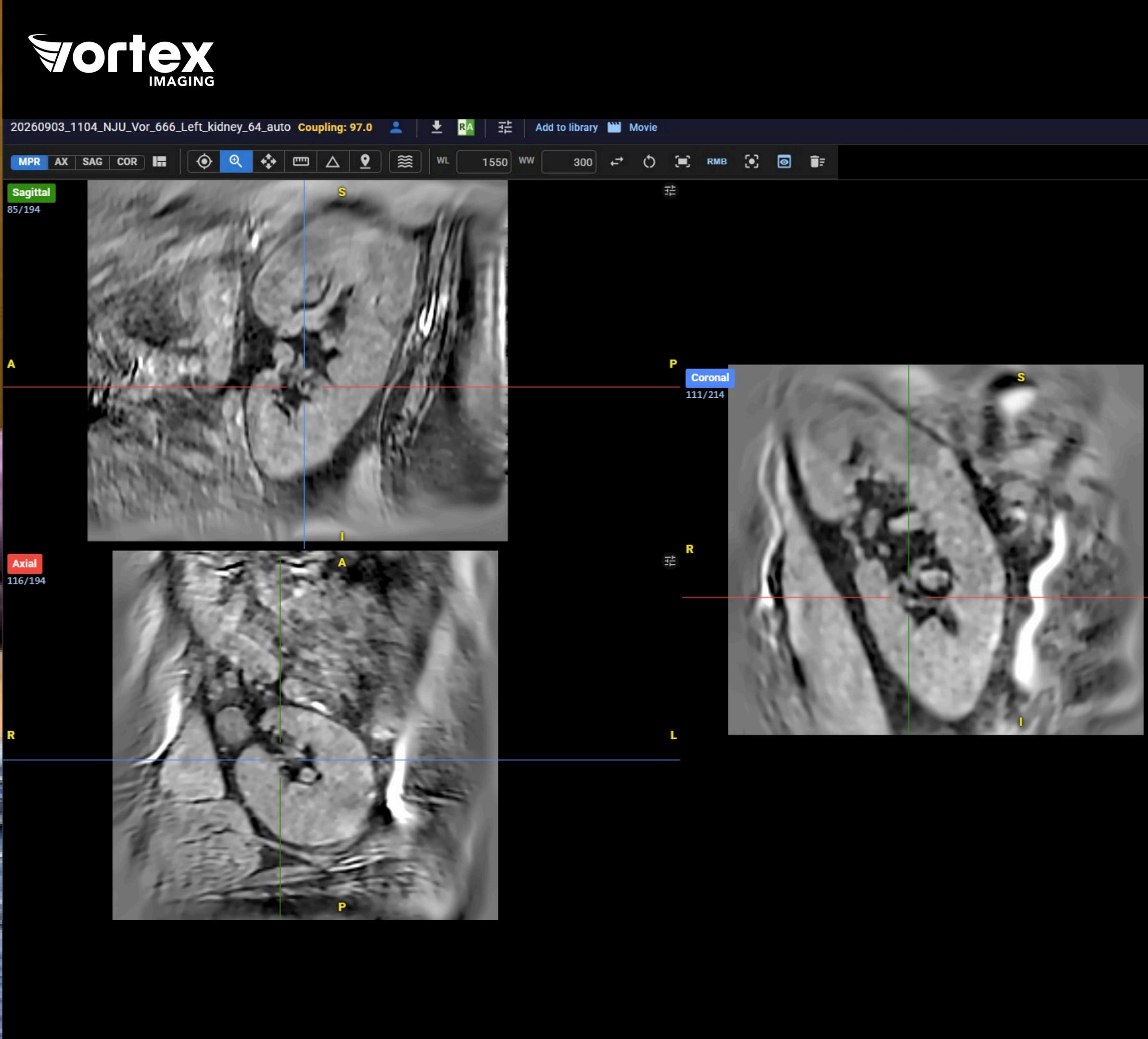
0 5

Configurable

Low-frequency, wide-aperture, full-aperture TX/RX with no beamforming: an unusual configuration, supported as standard.



One acquisition.
The full volume.
Stunning images



GPU Update / Refresher

